

Multiple Treatment Designs

AGRON 5130

YOUR NAME HERE

Introduction

The goal of this assignment is to give you hands-on practice working with **multiple treatment experimental designs** and their analyses. The exercises build directly on the concepts introduced in this week's lecture and lab.

You will analyze three agronomic experiments and practice:

- identifying experimental designs
- fitting the correct statistical models
- interpreting main effects and interactions
- visualizing treatment responses

Submit a **knitted PDF generated from your R Markdown file** to Canvas.

Question 1

1 point

Fifteen canola entries were compared in a field trial in Manitoba. The experiment was conducted using a **randomized complete block design with three blocks**. In this dataset block is coded as rep.

Load the dataset "data/canola_data.csv" and print the first six rows.

```
canola <- read.csv('./data/canola_data.csv')
head(canola)
```

```
##   rep entry yield
## 1  R1     A 33.17
## 2  R1     B 34.30
## 3  R1     C 35.84
## 4  R1     D 30.34
## 5  R1     E 33.91
## 6  R1     F 35.31
```

Question 2

2 points

Fit an appropriate model to evaluate the effect of **canola entry on yield** while accounting for block effects. Print the ANOVA table for your model.

```
model1 <- lm(yield ~ entry + rep,
             data = canola)
anova(model1)

## Analysis of Variance Table
##
## Response: yield
##          Df Sum Sq Mean Sq F value    Pr(>F)
## entry     14 1287.08   91.935   4.7954 0.0002086 ***
## rep        2  231.16  115.579   6.0287 0.0066476 **
## Residuals 28   536.80   19.172
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Question 3

2 points

Based on the ANOVA results from Question 2:

- Is there evidence that **canola entry affects yield**?
- Why is it important to include **block** in the model?

Write 2–3 sentences.

Question 4

1 point

Five barley genotypes were evaluated for their response to **three levels of drought stress** in an arid environment. The experiment was conducted as a **factorial treatment structure within a randomized complete block design**.

Load the dataset "data/barley_drought.csv" and print the first six rows.

```
barley <- read.csv('./data/barley_drought.csv')
head(barley)
```

```
##   block drought genotype    yield
## 1    R1    mild      HVU 786.1882
## 2    R1    mild      HSP 500.86027
## 3    R1    mild     HMUR 291.46258
## 4    R1    mild     HMAR  99.23831
## 5    R1    mild     OWHS 198.56284
## 6    R1 intense     HVU 538.66422
```

Question 5

2 points

Fit a factorial model that includes:

- block (you may treat it as fixed or random)
- genotype
- drought
- genotype $\times$ drought interaction

Print the ANOVA table.

```
model2 <- lm(yield ~ genotype * drought + block,
             data = barley)
anova(model2)

## Analysis of Variance Table
##
## Response: yield
##              Df Sum Sq Mean Sq  F value    Pr(>F)
## genotype      4 2450386   612596 158.6507 < 2.2e-16 ***
## drought       2   32175    16088   4.1664 0.0223548 *
## block         3    9676     3225   0.8353 0.4821208
## genotype:drought 8  140146    17518   4.5369 0.0004945 ***
## Residuals    42  162174     3861
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Question 6

2 points

Based on the ANOVA table from Question 5, interpret the results.

Address the following questions:

- Is drought stress a significant factor affecting yield?
- Do genotypes differ in yield?
- Is the **genotype:drought interaction** significant?

Question 7

1 point

In your own words, explain what a significant **genotype:drought** interaction means in the context of this experiment.

Question 8

1 point

The effects of **irrigation and tillage on soil nitrite (NO₂ emission)** were studied using a **split-plot design**, where irrigation was applied to main plots and tillage was applied to subplots.

Load the dataset "data/irrigation_tillage_no2.csv" and assign it to an object named `nitrite`. Print the first six rows.

```
nitrate <- read.csv('./data/irrigation_tillage_no2.csv')
head(nitrate)
```

```
##   block irrigation      tillage      NO2
## 1    B1  sprinkler conventional 0.8845244
## 2    B1  sprinkler      notill 0.7907164
## 3    B1    flood conventional 2.0050088
## 4    B1    flood      notill 1.9088118
## 5    B2  sprinkler conventional 0.5874839
## 6    B2  sprinkler      notill 1.0440447
```

Question 9

2 points

Fit an appropriate **split-plot model** for this experiment using a mixed-effects model and print the ANOVA table.

hint: there is a package that starts with `nl` and ends with `lme` that fits mixed-effects models.

```
library(nlme)
model3 <- lme(NO2 ~ tillage * irrigation,
              random = ~ 1 | block/irrigation,
              data = nitrate)
anova(model3)
```

```
##               numDF denDF    F-value p-value
## (Intercept)         1     6 1679.0345 <.0001
## tillage             1     6   2.4252  0.1704
## irrigation          1     3 303.3275  0.0004
## tillage:irrigation   1     6  25.2471  0.0024
```

Question 10

1 point

Based on the ANOVA table and the interaction plot, briefly explain whether irrigation, tillage, or their interaction affects NO₂ emissions.

Question 11 (Bonus)

2 points

This one is optional

In your own words, explain **why split-plot experiments require a different statistical model than completely randomized or randomized complete block designs.**